

Operations Research: Algebra and Proofs

Proof problem set for Lectures 1 to 5, with full solutions

The review sheet (notation, moves M1–M10, patterns P1–P11) is in
OR_Theory_Review_and_Proof_Set.pdf.

1 Problem Set: Algebra and Proofs

Rules. Give complete proofs. Any result of the textbook chapters may be quoted by name (for example “the tableau theorem”, “purification”, “strong duality”, “the outcome table”). A result proved in an earlier problem may be used in a later one. Unless stated otherwise, the setting is the equational form (E) with A of full row rank m , and (E*) is its dual $\min\{b^\top p : A^\top p \geq c\}$. Stars give the difficulty.

Every problem is new: none repeats Problem Sets 1 to 4 or the textbook’s examples. Each one practises the moves (M) and patterns (P) of the review.

Part A. The algebra of bases and tableaux

A1. The Lagrangian identity and the dual of the equational form

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- (a) Prove that for every $p \in \mathbb{R}^m$ and every $x \in \mathbb{R}^n$ with $Ax = b$,

$$c^\top x = b^\top p + (c - A^\top p)^\top x.$$

- (b) Use the dualization rules to show that the dual of (E) is (E*): $\min b^\top p$ s.t. $A^\top p \geq c$, p free. Then prove weak duality for this pair using only (a).
- (c) Let B be a basis and $p^\top = c_B^\top B^{-1}$. Show that $(A^\top p)_B = c_B$, so that $c - A^\top p$ is the reduced cost vector r and $r_B = 0$. Deduce the objective row of the tableau, $c^\top x = c_B^\top B^{-1}b + \sum_{j \in N} r_j x_j$, from (a).
- (d) Prove that B is an optimal basis ($B^{-1}b \geq 0$ and $r \leq 0$) if and only if its basic solution is feasible for (E) and p is feasible for (E*). Show that both are then optimal, with equal values.
- (e) Prove complementary slackness for this pair: feasible x and p are both optimal if and only if $x_j (A^\top p - c)_j = 0$ for every j . Explain why the basic solution and the multipliers of *any* basis satisfy these products, whether or not they are feasible.

Solution

(a) Because $Ax = b$, moving the transpose (M2) gives $b^\top p = (Ax)^\top p = x^\top A^\top p = (A^\top p)^\top x$. Hence

$$b^\top p + (c - A^\top p)^\top x = (A^\top p)^\top x + c^\top x - (A^\top p)^\top x = c^\top x.$$

No sign condition was used: the identity holds for every p and for every solution of $Ax = b$, feasible or not (pattern P6).

(b) (E) is a maximisation with equality rows and nonnegative variables. An equality row receives a free dual variable p_i (Rule 1). A variable $x_j \geq 0$ gives the dual row $(A^\top p)_j \geq c_j$ (Rule 2). The dual objective is $b^\top p$, minimised. This is (E*).

Weak duality: let x be feasible for (E) and p for (E*). Then $c - A^\top p \leq 0$ and $x \geq 0$, so $(c - A^\top p)^\top x \leq 0$ (M3), and (a) gives $c^\top x \leq b^\top p$.

(c) The columns of A with indices in B are the columns of the matrix B , so the B -entries of

$A^\top p$ are $(A^\top p)_{B(k)} = A_{B(k)}^\top p$, that is, $(A^\top p)_B = B^\top p$. With $p^\top = c_B^\top B^{-1}$,

$$B^\top p = (p^\top B)^\top = (c_B^\top B^{-1} B)^\top = c_B.$$

So $(c - A^\top p)_B = 0$. Moreover $(c - A^\top p)^\top = c^\top - c_B^\top B^{-1} A = r^\top$, the reduced costs. Putting this into (a), the basic terms vanish and $b^\top p = c_B^\top B^{-1} b$:

$$c^\top x = c_B^\top B^{-1} b + \sum_{j \in N} r_j x_j \quad \text{for every } x \text{ with } Ax = b.$$

This is part (b) of the tableau theorem, obtained without solving for x_B .

(d) Let x be the basic solution: $x_N = 0$, $x_B = B^{-1}b$. It satisfies $Ax = Bx_B = b$ (M1), and $x \geq 0$ if and only if $x_B \geq 0$. So $B^{-1}b \geq 0$ means exactly that x is feasible for (E). Next, $r \leq 0$ means $c - A^\top p \leq 0$, that is, $A^\top p \geq c$: exactly the feasibility of p for (E*).

The values are equal: $c^\top x = c_B^\top x_B = c_B^\top B^{-1}b = p^\top b$. Now apply (b) twice (pattern P9). For every feasible x' , $c^\top x' \leq b^\top p = c^\top x$, so x is optimal. For every feasible p' , $b^\top p' \geq c^\top x = b^\top p$, so p is optimal.

In words: *an optimal basis is one that is primal feasible and dual feasible at the same time.*

(e) For feasible x and p , rearrange (a):

$$b^\top p - c^\top x = (A^\top p - c)^\top x = \sum_{j=1}^n x_j (A^\top p - c)_j,$$

and every term is a product of two nonnegative numbers.

($\Leftarrow$) If every product is 0, then $b^\top p = c^\top x$, and both are optimal as in (d).

($\Rightarrow$) If both are optimal, then (E) has an optimal solution, so it is feasible and bounded. Strong duality (valid for every primal–dual pair) gives $b^\top p = c^\top x$. So the sum of nonnegative terms is 0, and every term is 0 (P2).

A basic pair always satisfies the products. For $j \in N$, $x_j = 0$. For $j \in B$, $(A^\top p - c)_j = -r_j = 0$ by (c). Every product is therefore zero, whatever the signs. This gives a clean description of the two simplex methods:

- the primal simplex keeps x feasible and the products zero, and works towards $r \leq 0$ (dual feasibility);
- the dual simplex keeps $r \leq 0$ and the products zero, and works towards $B^{-1}b \geq 0$ (primal feasibility).

Both stop when both are feasible, which is (d).

A2. The pivot in matrix form

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Let B be a basis, $j \in N$, $u = B^{-1}A_j$, and suppose $u_\ell \neq 0$. Let B' be B with column ℓ replaced by A_j , and write $w = u - e_\ell$.

(a) Show that $B^{-1}B' = F := I + w e_\ell^\top$.

(b) Show that F is invertible with $F^{-1} = I - \frac{1}{u_\ell} w e_\ell^\top$, and deduce $B'^{-1} = F^{-1}B^{-1}$.

(c) Describe what left multiplication by F^{-1} does to the rows of a matrix M . Explain why this is exactly a pivot of the simplex tableau on the entry u_ℓ .

(d) Show that the multipliers and reduced costs of B' are

$$p'^\top = p^\top + \frac{r_j}{u_\ell} e_\ell^\top B^{-1}, \quad r'^\top = r^\top - \frac{r_j}{u_\ell} e_\ell^\top B^{-1}A.$$

Check that $r'_j = 0$ and $r'_{B(\ell)} = -r_j/u_\ell$, and interpret the sign of $r'_{B(\ell)}$ in a simplex pivot.

- (e) Check (d) on the running example of Chapter 5, $\max 3x + 2y$ s.t. $x + y + s_1 = 4$, $2x + y + s_2 = 6$, starting from the basis (s_1, x) with y entering.

Solution

(a) Column $k \neq \ell$ of B' is $A_{B(k)}$, and $B^{-1}A_{B(k)} = e_k$ (M4). Column ℓ of B' is A_j , and $B^{-1}A_j = u$. So $B^{-1}B'$ is the identity with column ℓ replaced by u . The matrix $we_\ell^\top$ is zero except in column ℓ , where it equals $w = u - e_\ell$. Adding it to I replaces column ℓ of I by $e_\ell + u - e_\ell = u$. Hence $B^{-1}B' = I + we_\ell^\top = F$.

(b) Let $G = I - \frac{1}{u_\ell}we_\ell^\top$. The number $e_\ell^\top w = u_\ell - 1$ is a scalar, so

$$FG = I + we_\ell^\top - \frac{1}{u_\ell}we_\ell^\top - \frac{1}{u_\ell}w(e_\ell^\top w)e_\ell^\top = I + \left(1 - \frac{1}{u_\ell} - \frac{u_\ell - 1}{u_\ell}\right)we_\ell^\top = I.$$

For square matrices $FG = I$ implies $GF = I$, so $F^{-1} = G$. Then $B' = BF$ is a product of invertible matrices, hence a basis, and $B'^{-1} = F^{-1}B^{-1}$. (This is M9 again: $\det F = u_\ell$.)

(c) For any matrix M with m rows,

$$F^{-1}M = M - \frac{1}{u_\ell}(u - e_\ell)(e_\ell^\top M),$$

and $e_\ell^\top M$ is row ℓ of M , written M_ℓ . Reading row by row:

- row ℓ becomes $M_\ell - \frac{u_\ell - 1}{u_\ell}M_\ell = \frac{1}{u_\ell}M_\ell$: the pivot row is divided by the pivot;
- row $k \neq \ell$ becomes $M_k - u_k \cdot \frac{1}{u_\ell}M_\ell$: subtract u_k times the new pivot row.

Apply this to $M = B^{-1}[A \mid b]$, the rows of the system $B^{-1}Ax = B^{-1}b$. Column j of M is u , and $F^{-1}u = e_\ell$ (because $Fe_\ell = u$). So one Gauss-Jordan pivot on u_ℓ turns $B^{-1}[A \mid b]$ into $B'^{-1}[A \mid b]$, and makes column j a unit column. The dictionary form $x_B = B^{-1}b - B^{-1}A_N x_N$ is the same system with the nonbasic terms moved to the right.

(d) We have $c_{B'} = c_B + (c_j - c_{B(\ell)})e_\ell$, $c_{B'}e_\ell = c_j$, and $c_{B'}u = c_B^\top u + (c_j - c_{B(\ell)})u_\ell$. So

$$c_{B'}^\top w = c_B^\top u + (c_j - c_{B(\ell)})u_\ell - c_j.$$

Therefore

$$\begin{aligned} c_{B'}^\top F^{-1} &= c_{B'}^\top - \frac{1}{u_\ell}(c_{B'}^\top w)e_\ell^\top \\ &= c_B^\top + (c_j - c_{B(\ell)})e_\ell^\top - \frac{1}{u_\ell}(c_B^\top u - c_j)e_\ell^\top - (c_j - c_{B(\ell)})e_\ell^\top \\ &= c_B^\top + \frac{c_j - c_B^\top u}{u_\ell}e_\ell^\top = c_B^\top + \frac{r_j}{u_\ell}e_\ell^\top, \end{aligned}$$

since $r_j = c_j - c_B^\top B^{-1}A_j = c_j - c_B^\top u$. Multiplying on the right by B^{-1} :

$$p'^\top = c_{B'}^\top B'^{-1} = c_{B'}^\top F^{-1}B^{-1} = p^\top + \frac{r_j}{u_\ell}e_\ell^\top B^{-1}.$$

Then $r'^\top = c^\top - p'^\top A = r^\top - \frac{r_j}{u_\ell}e_\ell^\top B^{-1}A$.

Checks. $e_\ell^\top B^{-1}A_j = u_\ell$ gives $r'_j = r_j - r_j = 0$. $e_\ell^\top B^{-1}A_{B(\ell)} = e_\ell^\top e_\ell = 1$ gives $r'_{B(\ell)} = -r_j/u_\ell$.

For $k \neq \ell$, $e_\ell^\top B^{-1}A_{B(k)} = 0$, so $r'_{B(k)} = 0$, as M4 requires for B' .

Interpretation. In a simplex pivot $r_j > 0$ (an improving entering variable) and $u_\ell = -d_{B(\ell)} > 0$ (the ratio test only uses rows with $d_{B(\ell)} < 0$). So $r'_{B(\ell)} < 0$: bringing back the variable that just left would lower the objective. The formula for r' is the tableau rule “subtract r_j/u_ℓ times the pivot row from the objective row”.

(e) With $B = (s_1, x)$: $B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$, $c_B = (0, 3)$, so $p^\top = (0, \frac{3}{2})$ and $x_B = B^{-1}b = (1, 3)$. For y : $u = B^{-1}(1, 1) = (\frac{1}{2}, \frac{1}{2})$ and $r_y = 2 - p^\top(1, 1) = \frac{1}{2}$. The ratio test gives

$\min(1/\frac{1}{2}, 3/\frac{1}{2}) = 2$ in row $\ell = 1$, so s_1 leaves. Then

$$p'^\top = (0, \frac{3}{2}) + \frac{1/2}{1/2} (1, -\frac{1}{2}) = (1, 1), \quad r'_{s_1} = -\frac{1/2}{1/2} = -1,$$

which are the multipliers $p = (1, 1)$ and the objective row $z = 10 - s_1 - s_2$ of the optimal tableau.

A3. A degenerate vertex has several bases

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- Prove: if x is a *nondegenerate* BFS, exactly one basis has x as its basic solution.
- Show that the converse fails: give a degenerate BFS that belongs to exactly one basis.
- Let x be the BFS of B with $x_{B(\ell)} = 0$, and suppose some $j \in N$ has $u_\ell = (B^{-1}A_j)_\ell \neq 0$. Prove that B' , obtained by replacing $A_{B(\ell)}$ with A_j , is a basis with the *same* basic solution x . Is any sign condition on u_ℓ needed?
- Consider $\max 2x_1 + x_2$ s.t. $x_1 \leq 1$, $x_2 \leq 1$, $x_1 + x_2 \leq 2$, $x \geq 0$, with slacks s_1, s_2, s_3 .
 - Prove that $(x_1, x_2) = (1, 1)$ is optimal with a one-line certificate.
 - Show that it is a degenerate BFS, and find every basis that has it as its basic solution.
 - Write the objective rows of the tableaux of $\{x_1, x_2, s_3\}$ and $\{x_1, x_2, s_1\}$. Which one certifies optimality? What does this say about the optimality test at a degenerate vertex?

Solution

(a) Let B' be any basis whose basic solution is x . Then $x_i = 0$ for every $i \notin B'$, so $\text{supp}(x) \subseteq B'$. Nondegeneracy means $x_B > 0$, so $\text{supp}(x) = B$ has exactly m elements. Since $|B'| = m$ too, $B' = \text{supp}(x) = B$ as index sets.

(b) Take $A = I_2$ and $b = (1, 0)$. Then $P = \{(1, 0)\}$, the only basis is $\{1, 2\}$, and its BFS $(1, 0)$ is degenerate. (Any example with $n = m$ works: there is only one candidate basis.)

(c) B' is a basis. By A2(a)–(b), or M9, $B^{-1}B'$ is the identity with column ℓ replaced by u , with determinant $u_\ell \neq 0$. So $B' = B(B^{-1}B')$ is invertible.

Same basic solution. The nonbasic set of B' is $N' = (N \setminus \{j\}) \cup \{B(\ell)\}$. The point x is zero on $N \setminus \{j\}$ (nonbasic for B), at j (since $j \in N$), and at $B(\ell)$ (by hypothesis). So $x_{N'} = 0$ and $Ax = b$. By the uniqueness in the tableau theorem, a solution of $Ax = b$ that vanishes on N' is the basic solution of B' . So x is the BFS of B' .

Signs. None. The simplex method requires $u_\ell > 0$ only to keep the *next* point feasible after a positive step. Here the step is $\theta = 0$, so the point does not move: a *degenerate pivot*.

(d)(i) $2x_1 + x_2 = x_1 + x_1 + x_2 \leq 1 + 2 = 3$ (add the first and third constraints), and $(1, 1)$ has value 3. So it is optimal.

(d)(ii) In equational form the point is $(x_1, x_2, s_1, s_2, s_3) = (1, 1, 0, 0, 0)$, with $m = 3$ but only 2 positive entries: degenerate. By the argument of (a), any basis containing it must contain $\text{supp} = \{x_1, x_2\}$, plus one slack. All three choices are invertible (the columns of x_1, x_2 and any unit vector are independent), so $\{x_1, x_2, s_1\}$, $\{x_1, x_2, s_2\}$ and $\{x_1, x_2, s_3\}$ all have $(1, 1, 0, 0, 0)$ as their BFS.

(d)(iii) Basis $\{x_1, x_2, s_3\}$: $x_1 = 1 - s_1$, $x_2 = 1 - s_2$, $s_3 = 0 + s_1 + s_2$, and

$$z = 3 - 2s_1 - s_2.$$

All reduced costs are negative: this tableau certifies optimality.

Basis $\{x_1, x_2, s_1\}$: $x_2 = 1 - s_2$, $x_1 = 2 - x_2 - s_3 = 1 + s_2 - s_3$, $s_1 = 1 - x_1 = 0 - s_2 + s_3$, and

$$z = 2(1 + s_2 - s_3) + (1 - s_2) = 3 + s_2 - 2s_3.$$

Here $r_{s_2} = +1 > 0$, so this tableau does *not* certify optimality, although the point is optimal. The ratio test for s_2 is blocked by the row $s_1 = 0 - s_2 + s_3$, so $\theta_{\max} = 0$. Pivoting s_2 in and s_1 out is the degenerate exchange of (c). It reaches $\{x_1, x_2, s_2\}$, whose objective row $z = 3 - s_1 - s_3$ does certify.

Lesson. At a degenerate vertex the optimality test depends on which basis you hold. “ $r_N \leq 0$ ” is sufficient for optimality, and necessary only for nondegenerate BFSs (P7).

A4. Splitting a free variable ★

In an LP with a free variable x_1 , substitute $x_1 = u - v$ with $u, v \geq 0$, and write the result in equational form. The columns of u and v are A_1 and $-A_1$, with costs c_1 and $-c_1$.

- Prove that no basis contains both u and v .
- Deduce that every BFS has $uv = 0$, so that a vertex represents x_1 as $u = \max(x_1, 0)$, $v = \max(-x_1, 0)$.
- Prove that $r_u + r_v = 0$ for every basis.
- Deduce that at an optimal basis $r_u = r_v = 0$, that is, $(A^\top p)_1 = c_1$. Which dualization rule have you just recovered?

Solution

- $1 \cdot A_1 + 1 \cdot (-A_1) = 0$ is a nontrivial vanishing combination, so any set of columns containing both is linearly dependent, and its matrix is not invertible.
- A BFS is zero outside its basis. By (a) at least one of u, v is nonbasic, hence 0, so $uv = 0$. Now read off u and v from $x_1 = u - v$. If $x_1 > 0$, then $u = x_1 + v > 0$, so $v = 0$ and $u = x_1$. If $x_1 < 0$, then $v > 0$, so $u = 0$ and $v = -x_1$. If $x_1 = 0$, then $u = v$, and $uv = 0$ gives $u = v = 0$.
- $r_u = c_1 - p^\top A_1$ and $r_v = -c_1 - p^\top (-A_1) = -(c_1 - p^\top A_1) = -r_u$.
- At an optimal basis every reduced cost is ≤ 0 , so $r_u \leq 0$ and $-r_u = r_v \leq 0$. Hence $r_u = r_v = 0$, that is, $c_1 = p^\top A_1 = (A^\top p)_1$. This is the rule “a free primal variable gives a dual equality”: by A1(d), p is dual optimal, and its constraint for the free variable is tight.

Part B. Polyhedra and vertices

B1. Optimal sets are faces ★

Let $S \subseteq \mathbb{R}^n$ be convex, let $c \in \mathbb{R}^n$, and suppose $v^* = \max_{x \in S} c^\top x$ is attained. Let $S^* = \{x \in S : c^\top x = v^*\}$.

- Prove that S^* is convex.
- Prove: if $z = \lambda u + (1 - \lambda)w$ with $u, w \in S$, $0 < \lambda < 1$ and $z \in S^*$, then $u, w \in S^*$.
- Deduce that if S^* has two distinct points it has infinitely many, and that every extreme point of S^* is an extreme point of S .
- Apply this to an LP in equational form with a finite optimum to give a *second* proof that some optimal solution is a BFS.

Solution

- Let $x, y \in S^*$ and $0 \leq \lambda \leq 1$. Then $\lambda x + (1 - \lambda)y \in S$ by convexity, and $c^\top(\lambda x + (1 - \lambda)y) = \lambda v^* + (1 - \lambda)v^* = v^*$.
- $v^* = c^\top z = \lambda c^\top u + (1 - \lambda)c^\top w$, with $c^\top u \leq v^*$ and $c^\top w \leq v^*$. If either inequality were strict, then since $\lambda > 0$ and $1 - \lambda > 0$, we would get $c^\top z < \lambda v^* + (1 - \lambda)v^* = v^*$, a contradiction. So $c^\top u = c^\top w = v^*$.

(c) If $x \neq y$ are in S^* , the points $\lambda x + (1 - \lambda)y$ for $\lambda \in [0, 1]$ lie in S^* by (a), and they are pairwise distinct because $x - y \neq 0$. So there are infinitely many.

Let z be an extreme point of S^* , and suppose $z = \lambda u + (1 - \lambda)w$ with $u \neq w$ in S and $0 < \lambda < 1$. By (b), $u, w \in S^*$, so z is a proper combination of two distinct points of S^* . This contradicts extremality in S^* . So z is extreme in S .

(d) Let v^* be the optimal value of (E). Then $S^* = \{x : Ax = b, c^\top x = v^*, x \geq 0\}$ is nonempty and in equational form, with constraint matrix $\begin{bmatrix} A \\ c^\top \end{bmatrix}$. Delete a redundant row if this matrix is rank deficient; the solution set does not change. Maximise the objective 0 over S^* : it is feasible and bounded, so the fundamental theorem gives a BFS of S^* . By BFS = vertex, it is an extreme point of S^* . By (c) it is an extreme point of P , and by BFS = vertex again it is a BFS of (E). It lies in S^* , so it is optimal.

B2. A vertex is the unique optimum of some objective ★★

Prove that $\bar{x} \in P$ is a vertex of P if and only if there is $c \in \mathbb{R}^n$ such that $\bar{x}$ is the *unique* optimal solution of $\max\{c^\top x : x \in P\}$.

Solution

($\Leftarrow$) Suppose $\bar{x} = \lambda u + (1 - \lambda)w$ with $u, w \in P$ and $0 < \lambda < 1$. Since $\bar{x}$ is optimal, B1(b) shows that u and w are optimal too. By uniqueness, $u = w = \bar{x}$. So $\bar{x}$ is not a proper combination of two distinct points of P : it is a vertex.

($\Rightarrow$) By BFS = vertex, $\bar{x}$ is the BFS of some basis B . Define $c_j = 0$ for $j \in B$ and $c_j = -1$ for $j \in N$. For every $x \in P$,

$$c^\top x = - \sum_{j \in N} x_j \leq 0 = c^\top \bar{x},$$

so $\bar{x}$ is optimal. If $c^\top x = 0$, then the nonnegative numbers $x_j, j \in N$, sum to 0, so $x_N = 0$ (P2). By the tableau theorem (a), $x_B = B^{-1}b = \bar{x}_B$, so $x = \bar{x}$. Hence $\bar{x}$ is the unique optimum.

Tableau view. With $c_B = 0$ we get $p = 0$ and $r = c$, so $r_N = (-1, \dots, -1) < 0$, and the uniqueness corollary applies directly. Every vertex can be singled out by some objective, so no vertex is “useless”: the simplex method may have to visit any of them.

B3. A bounded P is the convex hull of its vertices ★★★

Assume $P = \{x : Ax = b, x \geq 0\}$ is nonempty and *bounded*.

(a) Let $x \in P$ have linearly dependent support columns, and let $d \neq 0$ satisfy $Ad = 0$ and $\text{supp}(d) \subseteq \text{supp}(x)$. Prove that d has at least one positive and at least one negative entry.

(b) Let $\theta_+ = \max\{\theta \geq 0 : x + \theta d \geq 0\}$ and $\theta_- = \max\{\theta \geq 0 : x - \theta d \geq 0\}$. Show that both are finite and positive, that $y = x + \theta_+ d$ and $z = x - \theta_- d$ lie in P and have smaller supports than x , and that

$$x = \frac{\theta_-}{\theta_+ + \theta_-} y + \frac{\theta_+}{\theta_+ + \theta_-} z.$$

(c) Prove by induction on $|\text{supp}(x)|$ that every $x \in P$ is a convex combination of BFSs.

(d) Deduce, *without* the fundamental theorem, that $\max\{c^\top x : x \in P\}$ is attained at a BFS.

(e) Show that (c) fails without boundedness.

Solution

(a) If $d \geq 0$, then for every $\theta \geq 0$ the point $x + \theta d$ satisfies $A(x + \theta d) = b$ and $x + \theta d \geq 0$, so it lies in P . Since $d \neq 0$, some $d_j > 0$, and $x_j + \theta d_j \rightarrow \infty$: P would be unbounded. So d has a negative entry. Applying the same argument to $-d$, which has the same properties, d also has

a positive entry.

(b) By M6,

$$\theta_+ = \min_{j: d_j < 0} \frac{x_j}{-d_j}, \quad \theta_- = \min_{j: d_j > 0} \frac{x_j}{d_j}.$$

Both minima are over nonempty sets by (a), so both are finite. Every index involved lies in $\text{supp}(d) \subseteq \text{supp}(x)$, where $x_j > 0$, so both are positive. Since $Ad = 0$, $Ay = Az = b$, and $y, z \geq 0$ by the choice of $\theta_{\pm}$, so $y, z \in P$.

Smaller supports. Outside $\text{supp}(x)$ both x and d vanish, so neither y nor z gains a new nonzero entry. The index attaining θ_+ becomes 0 in y , and the index attaining θ_- becomes 0 in z .

The combination. The weights are positive and sum to 1, and

$$\frac{\theta_-(x + \theta_+d) + \theta_+(x - \theta_-d)}{\theta_+ + \theta_-} = \frac{(\theta_+ + \theta_-)x + (\theta_- \theta_+ - \theta_+ \theta_-)d}{\theta_+ + \theta_-} = x.$$

(c) Induction on $k = |\text{supp}(x)|$ (P5). If the support columns of x are independent (this includes $k = 0$), then, as in the proof of BFS = vertex, they extend to a basis (rank m) and x is its BFS: a convex combination with a single term. Otherwise, (b) writes x as a convex combination of $y, z \in P$ with smaller supports. By the induction hypothesis, $y = \sum_i \alpha_i v_i$ and $z = \sum_i \beta_i v_i$ are convex combinations of BFSs v_i . Then

$$x = \mu y + (1 - \mu)z = \sum_i (\mu \alpha_i + (1 - \mu) \beta_i) v_i,$$

with nonnegative coefficients summing to $\mu + (1 - \mu) = 1$: a convex combination of BFSs.

(d) By (c) applied to any point of P , BFSs exist. There are finitely many, at most $\binom{n}{m}$, so one of them, v^* , has the largest value. For $x = \sum_i \lambda_i v_i \in P$,

$$c^\top x = \sum_i \lambda_i c^\top v_i \leq \sum_i \lambda_i c^\top v^* = c^\top v^*.$$

So v^* is optimal (P10). Boundedness of P replaced boundedness of the objective.

(e) Take $m = 1$, $n = 2$, $A = \begin{pmatrix} 1 & -1 \end{pmatrix}$, $b = 0$: $P = \{(t, t) : t \geq 0\}$, a ray. Its only BFS is $(0, 0)$ (the basis $\{1\}$ gives $x_1 = 0$; so does $\{2\}$). The point $(1, 1) \in P$ is not a convex combination of $\{(0, 0)\}$. Part (a) fails here because $d = (1, 1) \geq 0$.

B4. Carathéodory for cones

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Let $A \in \mathbb{R}^{m \times n}$ be *any* matrix (no rank assumption) and suppose $b = Ax$ for some $x \geq 0$.

- Prove that there is $x' \geq 0$ with $Ax' = b$ whose support columns $\{A_j : x'_j > 0\}$ are linearly independent. In particular, x' has at most $\text{rank } A \leq m$ positive entries.
- A chemist must make a mixture with exact composition $b \in \mathbb{R}^m$ (amounts of m chemicals) from n stock solutions, one litre of solution j supplying A_j . Show that if it can be done at all, it can be done with at most m stock solutions.
- Compare with purification: which hypothesis of purification is not needed here, and why?

Solution

(a) Among all $x \geq 0$ with $Ax = b$, choose one with the fewest positive entries; it exists because these numbers are nonnegative integers (P5, minimal counterexample). Suppose its support columns were dependent. Then there is $d \neq 0$ with $\text{supp}(d) \subseteq \text{supp}(x)$ and $Ad = 0$ (M7). Replacing d by $-d$ if necessary, d has a negative entry. Let $\theta^* = \min_{d_j < 0} x_j / (-d_j)$, which is positive since those j lie in $\text{supp}(x)$.

The point $x' = x + \theta^* d$ satisfies $Ax' = b$ and $x' \geq 0$ (M6). It has no new positive entries, and

the minimising index is 0, so it has fewer positive entries than x : a contradiction. Hence the support columns of the chosen x are independent. Independent vectors lie in the column space, of dimension rank A , so there are at most rank $A \leq m$ of them.

(b) A recipe is $x \geq 0$ (litres of each solution) with $Ax = b$. By (a), some recipe uses only solutions with independent composition vectors, at most m of them.

(c) Purification needs the objective to be bounded, so that a direction with $c^\top d \geq 0$ can be oriented to have a negative entry. Here there is no objective to protect, so either d or $-d$ will do, and every $d \neq 0$ has a negative entry or its negative does. Full row rank is also unnecessary: we never extend the support to a basis.

Part C. Simplex theory

C1. Cycling forces degeneracy

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The simplex method of Chapter 5 produces bases $B_0, B_1, \dots, B_k$ with $B_k = B_0$ and $k \geq 1$. Let x^t be the BFS of B_t , let j_t be the entering index at step t (so $r_{j_t} > 0$), and let $\theta_t \geq 0$ be the step length.

- Prove that $c^\top x^{t+1} - c^\top x^t = \theta_t r_{j_t} \geq 0$, and deduce that all the values $c^\top x^t$ are equal.
- Deduce that every $\theta_t = 0$ and that $x^0 = x^1 = \dots = x^k$.
- Prove that this common point is degenerate, in two ways: with the ratio test, and with Problem A3(a).
- Conclude that if every BFS is nondegenerate, the method cannot cycle.

Solution

(a) The step moves from x^t to $x^t + \theta_t d^{(j_t)}$, which is the BFS x^{t+1} of the exchanged basis. By the move theorem (c), its value is $c^\top x^t + \theta_t r_{j_t}$, and $\theta_t \geq 0$, $r_{j_t} > 0$ make the change nonnegative. So the values are nondecreasing along the cycle, and since $x^k = x^0$,

$$0 = c^\top x^k - c^\top x^0 = \sum_{t=0}^{k-1} \theta_t r_{j_t}.$$

This is a sum of nonnegative terms equal to zero, so every term is zero (P2), and all values are equal.

(b) Each term $\theta_t r_{j_t} = 0$ with $r_{j_t} > 0$ forces $\theta_t = 0$. Then $x^{t+1} = x^t + 0 \cdot d = x^t$ for every t .

(c) *Ratio test.* If x^0 were nondegenerate, every ratio $x_{B(k)} / (-d_{B(k)})$ would be positive, so $\theta_0 = \theta_{\max} > 0$, contradicting (b). (A finite θ_0 exists, since the method made a step.)

Problem A3(a). $B_1 \neq B_0$, since one index was exchanged, but by (b) both have the basic solution x^0 . A nondegenerate BFS has exactly one basis, so x^0 is degenerate.

(d) A cycle would produce a degenerate BFS by (c). So without degeneracy no basis repeats, and since there are at most $\binom{n}{m}$ bases, the method stops. This is the termination theorem again.

C2. When is the optimum unique?

★★

- Let x^* be an optimal BFS of the basis B , and assume x^* is *nondegenerate*. Prove that x^* is the unique optimal solution if and only if $r_j < 0$ for every $j \in N$.
- Show that the nondegeneracy hypothesis cannot be dropped: in the LP of Problem A3(d) with objective $\max x_1 + x_2$, show that $(1, 1)$ is the unique optimum but some optimal basis has a zero reduced cost.

Solution

(a) ($\Leftarrow$) is the uniqueness corollary of Chapter 5.

($\Rightarrow$) Since x^* is optimal and nondegenerate, $r_N \leq 0$ (the reduced-cost theorem, part 2). Suppose some $r_j = 0$ with $j \in N$, and let $d = d^{(j)}$. Nondegeneracy makes $\theta_{\max} > 0$; pick any θ with $0 < \theta \leq \theta_{\max}$ (for instance $\theta = 1$ if $\theta_{\max} = \infty$). Then $y = x^* + \theta d$ is feasible by the ratio test theorem, and has value $c^\top x^* + \theta r_j = c^\top x^*$: it is optimal. Also $y_j = \theta > 0 = x_j^*$, so $y \neq x^*$. This contradicts uniqueness. Hence $r_j < 0$ for every $j \in N$.

(b) $x_1 + x_2 \leq 2$ is a constraint, so the value is at most 2, attained at $(1, 1)$. If $x_1 + x_2 = 2$ with $x_1 \leq 1$ and $x_2 \leq 1$, then $x_1 = x_2 = 1$. So $(1, 1)$ is the unique optimum.

With the basis $\{x_1, x_2, s_1\}$ of A3(d), the rows are $x_1 = 1 + s_2 - s_3$, $x_2 = 1 - s_2$, so

$$z = x_1 + x_2 = 2 + 0 \cdot s_2 - s_3.$$

All reduced costs are ≤ 0 and the BFS is $(1, 1)$, so this basis is optimal, yet $r_{s_2} = 0$. Moving s_2 is blocked by the row $s_1 = 0 - s_2 + s_3$ ($\theta_{\max} = 0$), so the zero reduced cost does not produce a second optimum. The proof of (a) breaks exactly at “ $\theta_{\max} > 0$ ”.

C3. Certificates hidden in a tableau

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- (a) **Unboundedness.** At a BFS with basis B , suppose $r_j > 0$ and $d_B = -B^{-1}A_j \geq 0$. Show that $d = d^{(j)}$ satisfies $d \geq 0$, $Ad = 0$ and $c^\top d > 0$. Use this to prove *directly*, without the outcome table, that (E^*) is infeasible.
- (b) **Infeasibility.** A tableau row $x_{B(\ell)} = \beta_\ell + \sum_{j \in N} q_{\ell j} x_j$ has $\beta_\ell < 0$ and $q_{\ell j} \leq 0$ for every $j \in N$. Let $g^\top = e_\ell^\top B^{-1}$, row ℓ of B^{-1} . Show that $A^\top g \geq 0$ and $b^\top g < 0$, and prove directly that $P = \emptyset$.
- (c) In (b), assume also that every $r_j \leq 0$, as in the dual simplex method. Show that (E^*) is unbounded below, by exhibiting a ray of dual feasible points.

Solution

(a) By definition $d_j = 1$, $d_i = 0$ for the other nonbasic i , and $d_B \geq 0$ by hypothesis, so $d \geq 0$. $Ad = Bd_B + A_j = -A_j + A_j = 0$, and $c^\top d = c_j + c_B^\top d_B = c_j - c_B^\top B^{-1}A_j = r_j > 0$. Suppose p were feasible for (E^*) , so $A^\top p \geq c$. Since $d \geq 0$, M3 gives $(A^\top p)^\top d \geq c^\top d > 0$. But $(A^\top p)^\top d = p^\top(Ad) = 0$ (M2). This contradiction shows (E^*) is infeasible. In Farkas language (Problem D1(b)), d is the certificate that $A^\top p \geq c$ has no solution.

(b) By M5, $g^\top b = e_\ell^\top B^{-1}b = \beta_\ell < 0$. For the columns of A :

- $j \in N$: $g^\top A_j = (B^{-1}A_j)_\ell = -q_{\ell j} \geq 0$;
- $j = B(k)$: $g^\top A_{B(k)} = e_\ell^\top e_k \in \{0, 1\}$ (M4), which is ≥ 0 .

So $A^\top g \geq 0$. If some $x \in P$ existed, then $g^\top b = g^\top Ax = (A^\top g)^\top x \geq 0$, since $A^\top g \geq 0$ and $x \geq 0$. This contradicts $g^\top b < 0$, so $P = \emptyset$. So g is exactly a certificate of alternative (ii) of the equational Farkas lemma (Problem D1(a)), read off from one row of the tableau.

(c) If $r \leq 0$ then $p = B^{-\top} c_B$ is feasible for (E^*) (A1(d)). For $t \geq 0$,

$$A^\top(p + tg) = A^\top p + t A^\top g \geq A^\top p \geq c,$$

so $p + tg$ is dual feasible, and $b^\top(p + tg) = b^\top p + t \beta_\ell \rightarrow -\infty$. So (E^*) is unbounded below. This is consistent with the outcome table: an infeasible primal has an unbounded or infeasible dual, and here the dual is feasible.

Part D. Duality and theorems of the alternative

D1. Three faces of Farkas

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Take as known the inequality form: exactly one of (i) $\exists x: Ax \leq b$, and (ii) $\exists y \geq 0: A^\top y = 0, b^\top y < 0$, holds.

- (a) **Equational form.** Prove that exactly one of the following holds: (i) $\exists x \geq 0$ with $Ax = b$; (ii) $\exists y$ with $A^\top y \geq 0$ and $b^\top y < 0$.
- (b) **Dual-feasibility form.** Prove that exactly one of the following holds: (i) $\exists p$ with $A^\top p \geq c$; (ii) $\exists d \geq 0$ with $Ad = 0$ and $c^\top d > 0$.
- (c) For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $b = (-1, 1)$, decide which alternative of (a) holds and give the certificate. Interpret (a) geometrically.

Solution

(a) *Not both.* If $x \geq 0, Ax = b, A^\top y \geq 0$, then $b^\top y = (Ax)^\top y = (A^\top y)^\top x \geq 0$ (M2, M3), contradicting $b^\top y < 0$.

At least one. Suppose (i) fails. The conditions $Ax = b, x \geq 0$ form the inequality system

$$\begin{bmatrix} A \\ -A \\ -I \end{bmatrix} x \leq \begin{bmatrix} b \\ -b \\ 0 \end{bmatrix},$$

which then has no solution. The inequality form gives $(u, v, w) \geq 0$ with $A^\top u - A^\top v - w = 0$ and $b^\top u - b^\top v < 0$. Put $y = u - v$. Then $A^\top y = w \geq 0$ and $b^\top y < 0$: alternative (ii).

(b) *Not both.* If $A^\top p \geq c, d \geq 0, Ad = 0$, then $0 = p^\top Ad = (A^\top p)^\top d \geq c^\top d$, contradicting $c^\top d > 0$.

At least one. (i) is the system $(-A^\top)p \leq -c$. If it has no solution, the inequality form gives $d \geq 0$ with $(-A^\top)^\top d = -Ad = 0$ and $(-c)^\top d < 0$, that is, $Ad = 0$ and $c^\top d > 0$.

(c) Writing $b = \lambda_1(1, 0) + \lambda_2(1, 1)$ forces $\lambda_2 = 1$ and $\lambda_1 = -2 < 0$, so (i) fails. For the certificate, take $y = (1, 0)$: $A^\top y = (1, 1) \geq 0$ and $b^\top y = -1 < 0$.

Geometry. (i) says b lies in the cone generated by the columns of A . (ii) says some hyperplane $\{z : y^\top z = 0\}$ has every column on one side ($y^\top A_j \geq 0$) and b strictly on the other ($y^\top b < 0$). Here the line $z_1 = 0$ separates: both columns have $z_1 \geq 0$, while b has $z_1 = -1$.

D2. Farkas from strong duality

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Assume strong duality for every primal–dual pair, and that the dual of the dual is the primal. Prove the equational Farkas lemma D1(a) from these, by looking at the LP $\max\{0^\top x : Ax = b, x \geq 0\}$ and its dual. Why does this show that strong duality and Farkas are “the same theorem”?

Solution

Not both: as in D1(a).

At least one. Let (Q) be $\max\{0^\top x : Ax = b, x \geq 0\}$. By A1(b) with $c = 0$, its dual is (Q*) $\min\{b^\top y : A^\top y \geq 0\}$. Suppose (ii) fails: no y has $A^\top y \geq 0$ and $b^\top y < 0$. Then:

- (Q*) is feasible, since $y = 0$ satisfies $A^\top y \geq 0$;
- its objective is bounded below by 0, since every feasible y has $b^\top y \geq 0$.

Apply strong duality with (Q*) as the primal (a minimisation that is feasible and bounded). Its dual, which is (Q) because the dual of the dual is the primal, has an optimal solution. In particular (Q) is feasible: there is $x \geq 0$ with $Ax = b$, which is (i).

Same theorem. The course proved strong duality *from* Farkas (via Fourier–Motzkin), and this problem proves Farkas *from* strong duality. So each implies the other: both say that whatever is true about a system of linear inequalities can be certified by a nonnegative combination of its rows.

D3. A self-dual LP

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Let $K \in \mathbb{R}^{n \times n}$ be skew-symmetric ($K^\top = -K$), let $c \in \mathbb{R}^n$, and consider

$$(S) \quad \max c^\top x \quad \text{s.t.} \quad Kx \leq -c, \quad x \geq 0.$$

- Prove that $x^\top Kx = 0$ for every x .
- Write the dual of (S) and show that it is $\min\{-c^\top y : Ky \leq -c, y \geq 0\}$, that is, minus (S) itself.
- Prove that if (S) is feasible, its optimal value is 0, and every feasible x with $c^\top x = 0$ is optimal.
- Check (c) for $K = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $c = (-1, 1)$.

Solution

- (a) $x^\top Kx$ is a scalar, so it equals its transpose: $x^\top Kx = x^\top K^\top x = -x^\top Kx$. Hence $2x^\top Kx = 0$.
- (b) (S) is canonical with data $(K, -c, c)$, so its dual is $\min (-c)^\top y$ s.t. $K^\top y \geq c, y \geq 0$. Since $K^\top = -K$, the constraint is $-Ky \geq c$, that is, $Ky \leq -c$. So the dual minimises $-c^\top y$ over the *same* feasible set F as (S), and $\min_F (-c^\top y) = -\max_F c^\top y$.
- (c) *Upper bound.* For $x \in F$, multiply $Kx \leq -c$ by $x \geq 0$ (M3): $x^\top Kx \leq -x^\top c$. By (a) this reads $0 \leq -c^\top x$, so $c^\top x \leq 0$ on F .
- Value 0.* (S) is feasible and bounded above, so it has an optimal value p^* , and by strong duality the dual has the same value. By (b) the dual value is $-p^*$. So $p^* = -p^*$, that is, $p^* = 0$. Any feasible x with $c^\top x = 0$ attains the bound and is optimal.
- (d) $Kx = (x_2, -x_1) \leq (1, -1)$ means $x_2 \leq 1$ and $x_1 \geq 1$. On this set $c^\top x = -x_1 + x_2 \leq -1 + 1 = 0$, with equality only at $(1, 1)$. So the optimum is 0, attained at $(1, 1)$, as (c) predicts.

D4. When is P bounded?

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Let $P = \{x : Ax = b, x \geq 0\}$ be nonempty. Prove that P is bounded if and only if there is no $d \in \mathbb{R}^n$ with $d \geq 0, d \neq 0$ and $Ad = 0$. *Hint for ($\Leftarrow$): maximise each coordinate x_j over P and use D1(b).*

Solution

($\Rightarrow$) Contrapositive. If such a d exists, take $x \in P$. For every $\theta \geq 0$, $A(x + \theta d) = b$ and $x + \theta d \geq 0$, so $x + \theta d \in P$. Some $d_j > 0$, and then $x_j + \theta d_j \rightarrow \infty$. So P is unbounded.

($\Leftarrow$) Assume no such d exists, and fix j . Consider the LP $\max\{x_j : x \in P\}$, that is, (E) with $c = e_j$. Its dual is $\min\{b^\top p : A^\top p \geq e_j\}$.

- If $A^\top p \geq e_j$ had no solution, D1(b) with $c = e_j$ would give $d \geq 0$ with $Ad = 0$ and $d_j > 0$. Such a d is nonzero, which is excluded. So the dual has a feasible $p^{(j)}$.
- By weak duality, every $x \in P$ satisfies $x_j \leq b^\top p^{(j)} =: M_j$.

Together with $x \geq 0$, every $x \in P$ lies in the box $0 \leq x_j \leq M_j$ for $j = 1, \dots, n$. So P is bounded.

Remark. With B3, this gives a complete picture: P is bounded exactly when $\{d \geq 0 : Ad = 0\} = \{0\}$, and then P is the set of convex combinations of its vertices.

Part E. Value functions and sensitivity

E1. The value function $v(b)$

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For fixed A and c , let $P(b) = \{x : Ax = b, x \geq 0\}$, $F = \{b : P(b) \neq \emptyset\}$, and $v(b) = \max\{c^\top x : x \in P(b)\}$.

- (a) Prove: if $v(b_0)$ is finite for one b_0 , then $v(b)$ is finite for every $b \in F$. *Hint: does the feasibility of (E^*) depend on b ?* Assume this from now on.
- (b) Prove that F is a convex cone: $b_1, b_2 \in F$ and $t \geq 0$ imply $b_1 + b_2 \in F$ and $tb_1 \in F$.
- (c) Prove that $v(tb) = tv(b)$ for $t > 0$, and $v(b_1 + b_2) \geq v(b_1) + v(b_2)$.
- (d) Deduce that v is concave on F .
- (e) Show that $v(0) = 0$. Interpret.

Solution

(a) Since $v(b_0)$ is finite, (E) at b_0 is feasible and bounded, so strong duality gives an optimal, in particular feasible, p_0 with $A^\top p_0 \geq c$. The condition $A^\top p \geq c$ does not involve b , so p_0 is dual feasible for every b . For $b \in F$, weak duality bounds every $x \in P(b)$ by $c^\top x \leq b^\top p_0$. A feasible LP with bounded objective has an optimal solution (fundamental theorem), so $v(b)$ is finite.

So for a given (A, c) , either v is finite on all of F or nowhere.

(b) Let $x_1 \in P(b_1)$ and $x_2 \in P(b_2)$. Then $x_1 + x_2 \geq 0$ and $A(x_1 + x_2) = b_1 + b_2$, so $b_1 + b_2 \in F$. For $t \geq 0$, $tx_1 \geq 0$ and $A(tx_1) = tb_1$, so $tb_1 \in F$. A set closed under addition and nonnegative scaling is convex, since $\lambda b_1 + (1 - \lambda)b_2 = (\lambda b_1) + ((1 - \lambda)b_2)$.

(c) *Homogeneity.* For $t > 0$, the map $x \mapsto tx$ is a bijection from $P(b)$ onto $P(tb)$, with inverse $x \mapsto x/t$, and it multiplies the objective by t (M10). So $v(tb) = tv(b)$.

Superadditivity. By (a) and the fundamental theorem there are optimal $x_1 \in P(b_1)$ and $x_2 \in P(b_2)$. The point $x_1 + x_2 \in P(b_1 + b_2)$ has value $v(b_1) + v(b_2)$, so $v(b_1 + b_2) \geq v(b_1) + v(b_2)$ (P11).

(d) For $0 < \lambda < 1$, superadditivity and then homogeneity give

$$v(\lambda b_1 + (1 - \lambda)b_2) \geq v(\lambda b_1) + v((1 - \lambda)b_2) = \lambda v(b_1) + (1 - \lambda)v(b_2).$$

The cases $\lambda \in \{0, 1\}$ are trivial.

(e) $0 \in F$ (take $x = 0$), so $v(0)$ is finite by (a), and $v(0) = v(2 \cdot 0) = 2v(0)$, so $v(0) = 0$. With no resources, the best plan is worth nothing, unless the LP is unbounded, in which case it is unbounded for every $b \in F$.

E2. Shadow prices are supergradients

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Keep the notation of E1, fix $b \in F$, and let p^* be an optimal solution of (E^*) at b .

- (a) Prove that $v(b') \leq v(b) + p^{*\top}(b' - b)$ for every $b' \in F$.
- (b) Deduce that for $\delta > 0$ (with $b \pm \delta e_i \in F$)

$$\frac{v(b + \delta e_i) - v(b)}{\delta} \leq p_i^* \leq \frac{v(b) - v(b - \delta e_i)}{\delta}.$$

- (c) In the running example of Chapter 5 ($\max 3x + 2y, x + y \leq 4 + \delta, 2x + y \leq 6$), the value is $v = 10 + \delta$ on $[-1, 2]$ and $v = 12$ for $\delta \geq 2$. At $\delta = 2$, use (b) to show that every optimal dual solution has $0 \leq p_1 \leq 1$, and exhibit two optimal dual solutions with $p_1 = 1$ and $p_1 = 0$.
- (d) State and prove the analogue of (a) for $w(c) = \max\{c^\top x : x \in P\}$, with an optimal x^* in the role of p^* .

Solution

(a) p^* satisfies $A^\top p^* \geq c$, which does not involve b , so it is dual feasible at b' as well. Weak duality at b' gives $v(b') \leq b'^\top p^*$. By strong duality at b , $b^\top p^* = v(b)$. So

$$v(b') \leq b^\top p^* + (b' - b)^\top p^* = v(b) + p^{*\top}(b' - b).$$

Geometrically, the graph of the concave function v lies below the hyperplane through $(b, v(b))$ with slope p^* .

(b) Take $b' = b + \delta e_i$: $v(b + \delta e_i) - v(b) \leq \delta p_i^*$. Divide by $\delta > 0$. Take $b' = b - \delta e_i$: $v(b - \delta e_i) - v(b) \leq -\delta p_i^*$, that is, $p_i^* \leq (v(b) - v(b - \delta e_i))/\delta$.

(c) At $\delta = 2$, for small $h > 0$: the right quotient is $(12 - 12)/h = 0$ and the left quotient is $(12 - (12 - h))/h = 1$. By (b), $0 \leq p_1 \leq 1$ for *every* optimal dual solution.

The right-hand side is $b = (6, 6)$, and the dual is $\min 6p_1 + 6p_2$ s.t. $p_1 + 2p_2 \geq 3$, $p_1 + p_2 \geq 2$, $p \geq 0$.

- $p = (1, 1)$: $(3, 2) \geq (3, 2)$ and value 12. It comes from the basis (x, y) , whose BFS $(0, 6)$ is degenerate at $\delta = 2$.
- $p = (0, 2)$: $(4, 2) \geq (3, 2)$ and value 12. It comes from the basis (s_1, y) .

Both have value $12 = v$, so both are optimal, and so is every point of the segment between them. The dual optimum is not unique because the primal optimum is degenerate (compare Part F, statement 9).

(d) **Claim:** $w(c') \geq w(c) + x^{*\top}(c' - c)$ for every c' . Indeed $x^* \in P$ is feasible for every objective, so $w(c') \geq c'^\top x^* = c^\top x^* + (c' - c)^\top x^* = w(c) + x^{*\top}(c' - c)$. So x^* is a subgradient of the convex function w , just as p^* is a supergradient of the concave function v .

E3. Optimality regions are convex

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- Fix an optimal basis B at (b, c) . Prove that $R = \{\Delta b : B \text{ is optimal for } b + \Delta b\}$ is a convex polyhedron containing 0, and that v is affine on $b + R$.
- Prove that $C_B = \{c : B \text{ is an optimal basis for } c\}$, with A and b fixed and $B^{-1}b \geq 0$, is a convex cone described by linear inequalities in c .
- Fix a feasible point x^* , not necessarily basic. Prove from the definition of optimality alone that $C(x^*) = \{c : x^* \text{ is optimal for } \max_{x \in P} c^\top x\}$ is a convex cone.
- For $\max c_1x + c_2y$ s.t. $x + y \leq 4$, $2x + y \leq 6$, $x, y \geq 0$, prove that $(2, 2)$ is optimal if and only if $c = p_1(1, 1) + p_2(2, 1)$ for some $p \geq 0$.

Solution

(a) The reduced costs do not depend on b , so B is optimal for $b + \Delta b$ exactly when $B^{-1}(b + \Delta b) \geq 0$. Hence $R = \{\Delta b : B^{-1}\Delta b \geq -B^{-1}b\}$, an intersection of finitely many half-spaces: a convex polyhedron. It contains 0 because $B^{-1}b \geq 0$. On $b + R$ the value is $c_B^\top B^{-1}(b + \Delta b) = v(b) + p^\top \Delta b$, an affine function of Δb .

(b) Since $B^{-1}b \geq 0$ does not involve c , B is optimal for c exactly when $r_N = c_N - (B^{-1}A_N)^\top c_B \leq 0$. Each entry is a linear function of c , so C_B is the solution set of finitely many homogeneous linear inequalities. If $c, c' \in C_B$ and $\lambda, \mu \geq 0$, the reduced costs of $\lambda c + \mu c'$ are $\lambda r_N + \mu r'_N \leq 0$. So C_B is a convex cone.

(c) Let $c, c' \in C(x^*)$ and $\lambda, \mu \geq 0$. For every $x \in P$,

$$(\lambda c + \mu c')^\top x = \lambda c^\top x + \mu c'^\top x \leq \lambda c^\top x^* + \mu c'^\top x^* = (\lambda c + \mu c')^\top x^*,$$

using $\lambda, \mu \geq 0$ (P11). So $\lambda c + \mu c' \in C(x^*)$.

(d) Let $A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ and $b = (4, 6)$. At $x^* = (2, 2)$ both constraints are tight, $Ax^* = b$, and both coordinates are positive.

($\Leftarrow$) If $c = A^\top p$ with $p \geq 0$, then p is feasible for (D) (it satisfies $A^\top p \geq c$ with equality), and $b^\top p = (Ax^*)^\top p = (A^\top p)^\top x^* = c^\top x^*$. Matching values make x^* optimal (P9).

($\Rightarrow$) If x^* is optimal, strong duality gives an optimal $y \geq 0$, and complementary slackness with $x_1^*, x_2^* > 0$ forces both dual constraints to be tight: $A^\top y = c$. So $c = y_1(1, 1) + y_2(2, 1)$ with $y \geq 0$.

Hence $C((2, 2))$ is the cone spanned by the normals of the two binding constraints. This is the picture of Chapter 5: the vertex is optimal while the objective vector stays between the two normals.

Part F. True or false? Justify each answer

F. Eighteen statements ★

1. Every nonempty polyhedron $\{x \in \mathbb{R}^n : Ax \leq b\}$ has at least one vertex.
2. Every nonempty $P = \{x : Ax = b, x \geq 0\}$ with $\text{rank } A = m$ has at least one vertex.
3. Every BFS has exactly m positive entries.
4. If $x \in P$ has at most m positive entries, then x is a BFS.
5. Two different bases always have different basic solutions.
6. In a maximisation, if a BFS has a reduced cost $r_j > 0$, it is not optimal.
7. If an LP has a unique optimal solution, every optimal basis has $r_N < 0$.
8. The reduced costs of a basis do not depend on the sequence of pivots that reached it.
9. If x^* is a nondegenerate optimal BFS of (E), then (E^*) has a unique optimal solution.
10. If (P) and (D) are both feasible, then both have optimal solutions.
11. If (P) is unbounded, then (D) is infeasible.
12. If (D) is infeasible, then (P) is unbounded.
13. If $Ax = b, x \geq 0$ has no solution, there is y with $A^\top y \geq 0$ and $b^\top y < 0$.
14. The optimal value $v(b)$ of a maximisation LP is a convex function of b .
15. A shadow price p_i gives the change in the optimal value for any change of b_i .
16. Changing the profit c_j of a nonbasic variable does not change the optimal value, as long as B stays optimal.
17. Adding a new variable to a maximisation LP can never decrease its optimal value.
18. If $v(b)$ is finite for one right-hand side b , the LP is not unbounded for any other right-hand side.

Solution

1. **False.** $\{x \in \mathbb{R}^2 : x_1 + x_2 \leq 1\}$: every point x is the midpoint of $x \pm (1, -1)$, both feasible because $(1, 1)^\top (1, -1) = 0$. So there is no vertex.
2. **True.** Purification with $c = 0$ (bounded objective) gives a BFS, which is a vertex.
3. **False.** Degenerate BFSs have fewer, e.g. $(1, 1, 0, 0, 0)$ in A3(d), with $2 < 3$.

4. **False.** $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ (rank 2), $b = (3, 3)$, $x = (1, 1, 0)$ has $2 = m$ positive entries, but its support columns $(1, 1)$ and $(2, 2)$ are dependent. With $d = (2, -1, 0)$, $Ad = 0$, so x is the midpoint of $x \pm \varepsilon d$ and is not a vertex.
5. **False.** In A3(d), three bases share the BFS $(1, 1, 0, 0, 0)$.
6. **False.** It needs nondegeneracy: the basis $\{x_1, x_2, s_1\}$ in A3(d) has $r_{s_2} = 1 > 0$ at an optimal point.
7. **False.** C2(b): unique optimum $(1, 1)$, optimal basis with $r_{s_2} = 0$.
8. **True.** $r = c - A^\top p$ with $p = B^{-\top} c_B$ depends only on B ; the tableau of a basis is unique.
9. **True.** Every optimal p satisfies complementary slackness with x^* (A1(e)). Since $x_B^* > 0$, this forces $(A^\top p)_B = c_B$, that is, $B^\top p = c_B$. So $p = B^{-\top} c_B$ is the only possibility.
10. **True.** Weak duality bounds (P), so it has an optimal solution, and strong duality gives one for (D).
11. **True.** A feasible y would give the bound $c^\top x \leq b^\top y$.
12. **False.** Both can be infeasible: (P) $\max 2x_1 - x_2$ s.t. $x_1 - x_2 \leq 1$, $-x_1 + x_2 \leq -2$, $x \geq 0$ (adding the rows gives $0 \leq -1$). Its dual has $y_1 - y_2 \geq 2$ and $-y_1 + y_2 \geq -1$ (adding gives $0 \geq 1$).
13. **True.** This is D1(a).
14. **False.** It is concave (E1). In the running example the slopes are $3, 1, 0$, decreasing.
15. **False.** Only on the range where the basis stays optimal. In the running example $p_1 = 1$, but beyond $\delta = 2$ the slope is 0.
16. **True.** The value $c_B^\top B^{-1} b$ does not involve c_N .
17. **True.** The old optimal solution, extended by 0 in the new variable, stays feasible with the same value.
18. **True.** By E1(a), v is finite on every feasible right-hand side, because the dual feasible set does not depend on b .